$$\begin{cases} x = -\dfrac{2\sqrt{3}-\sqrt{5}}{6} l \\ y = -\dfrac{2+\sqrt{15}}{6} l \end{cases}。 \qquad (25)$$

Let θ_2 be the rod's angle to the horizontal when ball 2 leaves (see figure). Then

$$\cos\theta_2 = \frac{|x|}{l} = \frac{2\sqrt{3}-\sqrt{5}}{6}, \qquad (26)$$

$$\theta_2 = \arccos\left(\frac{2\sqrt{3}-\sqrt{5}}{6}\right) = 78.2° \text{ (or } 1.36 \text{ rad)}. \qquad (27)$$

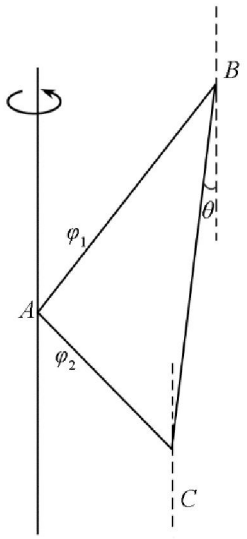
Scoring: (26) (15) ; ; each ; ; ;

(20) (27) ; ; each ; each .

IV. (15 points)

Method 1:

Let m be B's mass, θ the angle of BC to the vertical, and ω its angular speed. For pair BC, repulsion is f_{BC}. Newton's law gives



$$f_{AB}\sin\varphi_1 - f_{BC}\sin\theta = m\omega^2|AB|\sin\varphi_1, \qquad (1)$$

$$f_{AB}\cos\varphi_1 - f_{BC}\cos\theta = 0, \qquad (2)$$

$$f_{AC}\sin\varphi_2 + f_{BC}\sin\theta = m\omega^2|AC|\sin\varphi_2, \qquad (3)$$

$$f_{AC}\cos\varphi_2 - f_{BC}\cos\theta = 0。 \qquad (4)$$

(2)、(4) Using in (1)、(3) gives

$$\frac{f_{AB}\sin(\varphi_1-\theta)}{f_{AC}\sin(\varphi_2+\theta)} = \frac{|AB|\sin\varphi_1}{|AC|\sin\varphi_2}。 \qquad (5)$$

The geometry gives

$$\frac{|AB|}{AC} = \frac{\sin(\varphi_2+\theta)}{\sin(\varphi_1-\theta)} \qquad (6)$$

With the given expressions for f_{AB} and f_{BC}, (5) gives

$$\left(\frac{|AB|}{|AC|}\right)^{a-2} = \frac{\sin\varphi_1}{\sin\varphi_2} \qquad (7)$$

Also, (2)、(4) give $\dfrac{f_{AB}}{f_{AC}} = \dfrac{\cos\varphi_2}{\cos\varphi_1}$。 (8)

Substituting the given expressions for f_{AB} and f_{BC} gives

$$\frac{|AB|^{\alpha}}{|AC|^{\alpha}} = \frac{\cos\varphi_2}{\cos\varphi_1}。 \qquad (9)$$